

Worksheet 6A: Statistics and Probability — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A: Central Tendency

1. Mean: 85.71, Median: 85, Mode: 85
2. Mean: 145, Median: 145, Mode: 145
3. (a) Mean: 2278.57, Median: 1450, Mode: none (b) Mean (c) Median
4. (a) With: 570, Without: 298.33 (b) With: 310, Without: 300
5. (a) 2875 (b) 3000 (c) 5000 (d) Model may overfit to majority classes
6. (a) 2.998 (b) Exactly matches!
7. 85.55
8. 0.56

Part B: Spread

9. (a) 8 (b) -4, -2, 0, 2, 4 (c) 16, 4, 0, 4, 16 (d) 8 (e) 2.83
10. (a) Both 86% (b) A: 1.58, B: 10.30 (c) Model A
11. (a) Version 1.0: 1.25s (b) Version 1.0: 0.11, Version 2.0: 0.58 (c) Version 1.0
12. (a) 85-115 (b) 70-130 (c) 2.5%
13. (a) 168-182 cm (b) 189 cm (c) 1.5%

Part C: Probability

14. (a) $1/4$ (b) $1/13$ (c) $1/52$ (d) $40/52 = 10/13$
15. (a) $1/2$ (b) $1/2$ (c) $4/5$
16. (a) 0.80 (b) 0.15 (c) 0.05
17. (a) 0.90 (b) 0.85 (c) 0.80 (d) 0.86
18. (a) 100 sick, 9900 healthy (b) 95 (c) 990 (d) $95/(95+990) \approx 8.8\%$
19. (a) 0.30 (b) 0.9412 (c) ≈ 0.364

20. (a) 0.857 (b) 0.143 (c) 0.904

21. (a) 0.893 (b) ≈ 0.99997

Part D: Counting

22. $26 \times 10 \times 26 \times 10 = 67,600$

23. $4 \times 6 \times 3 = 72$

24. $5! = 120$

25. $P(5, 3) = 60$

26. $P(8, 3) = 336$

27. $C(10, 3) = 120$

28. $C(52, 5) = 2,598,960$

29. $C(49, 6) = 13,983,816$

30. (a) 120 (b) 252 (c) 1024

31. $3 \times 4 \times 4 \times 5 = 240$

32. (a) $4^3 = 64$ (b) More than enough to code 20 amino acids (with redundancy)

33. (a) $C(8, 2) = 28$ (b) 56 hours

Part E: ML Applications

34. (a) 0.965 (b) 0.81 (c) 0.85 (d) Low recall

35. (a) €5 (b) €0 (c) Fair game

36. (a) €30 (b) €3000

37. (a) 0.032 (b) 0.375

38. 800 fish

39. (a) 87% (b) 2.12% (c) $87\% \pm 2.12\%$

40. (a) 30% relative improvement

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